

Blood Glucose Homeostasis: Control of Blood glucose

Sources of Blood Glucose

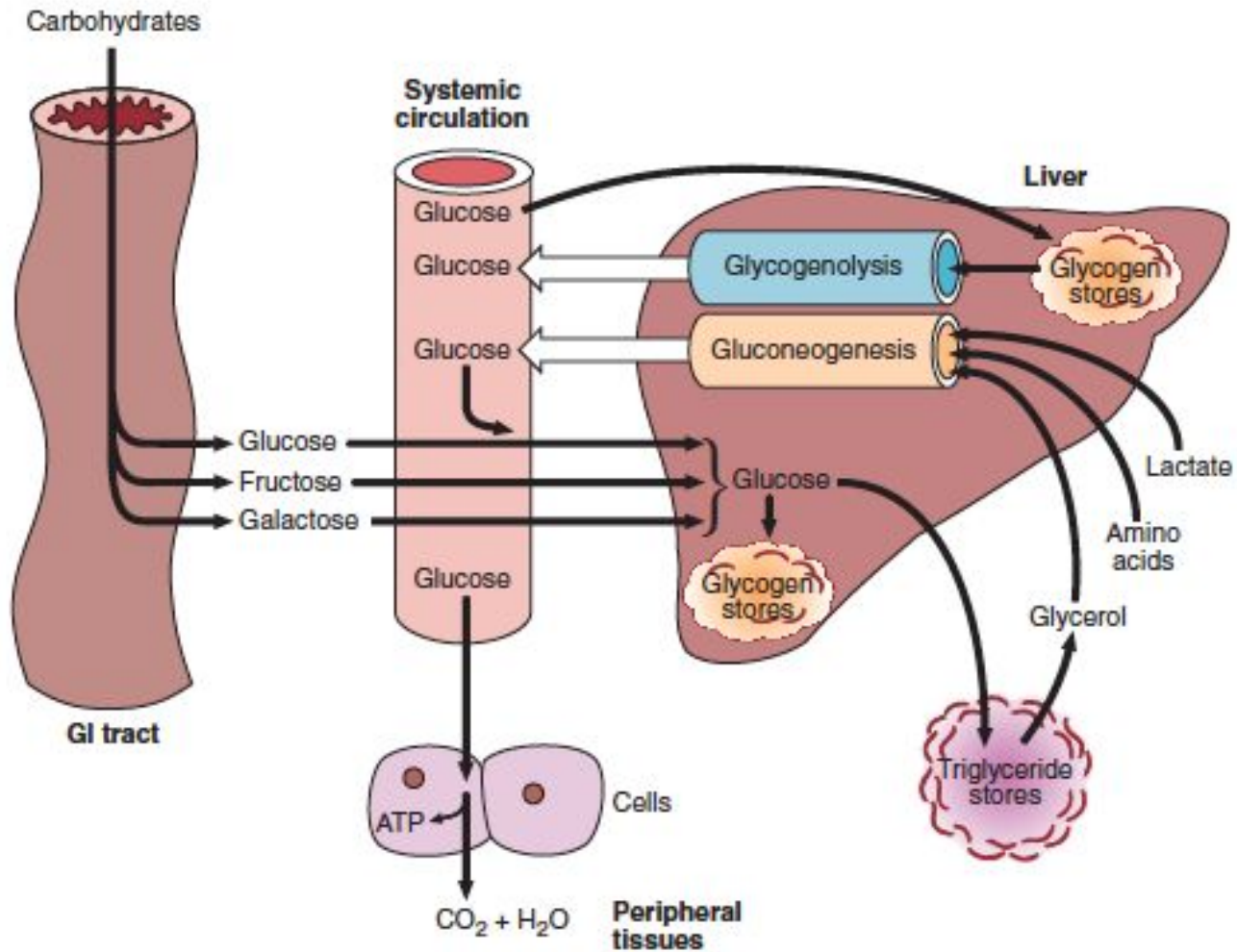
- Dietary carbohydrates:
 - Mainly glucose, galactose and fructose are produced from dietary carbohydrates.
 - They reach liver via portal vein. Galactose and fructose are also converted to glucose in liver.
 - Dietary carbohydrate maintains blood glucose level up to 3 hours after food intake.
- Liver glycogenolysis:
 - Maintains blood sugar level up to 16 hours after food intake.
- Gluconeogenesis:
 - Maintains blood sugar level up to 36 hours after food intake.
 - Beyond that period (If food is not taken or fasting) also gluconeogenesis is the only source of glucose.

Removal of Blood Glucose

Various pathways use glucose for different purposes.

- Glycolysis uses glucose for energy.
- Glycogenesis uses glucose for glycogen formation.
- HMP shunt and uronic acid pathways convert glucose to pentoses and mucopolysaccharides.
- Glucose is converted to fat also.

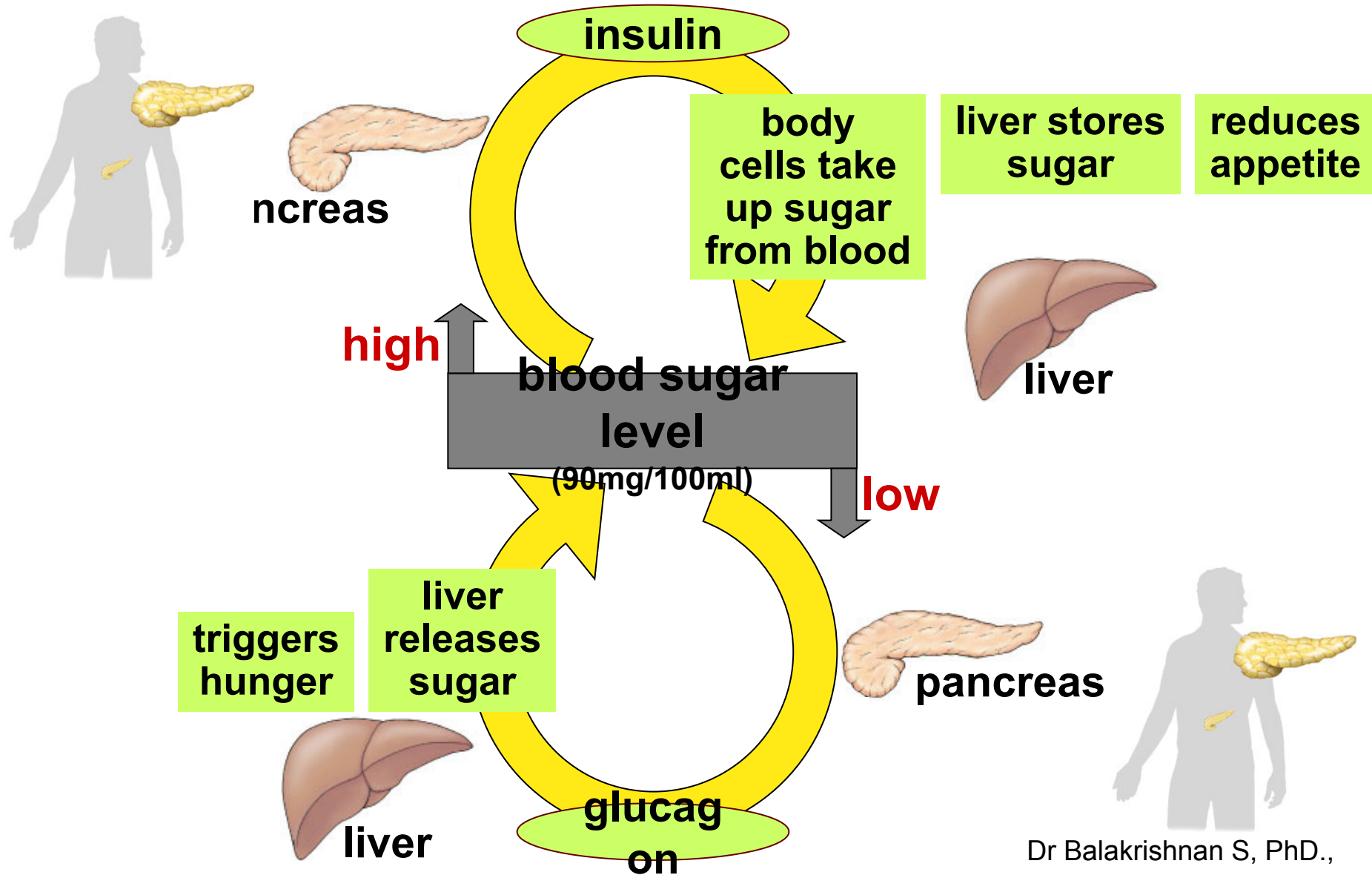
Glucose homeostasis



BLOOD GLUCOSE LEVEL AND ITS REGULATION

- Normal blood glucose level = 70-110 mg/dl.
- Above the normal □ hyperglycemia,
- Below the normal □ hypoglycemia.
- After the ingestion of a carbohydrate meal, it may rise to 120-140 mg/dl.
- Factors involved in the homeostasis (regulation) of blood glucose are:
 - Hormones
 - Metabolic processes
 - Renal mechanism.

Regulation of Blood Sugar



Regulation of Plasma Glucose Level

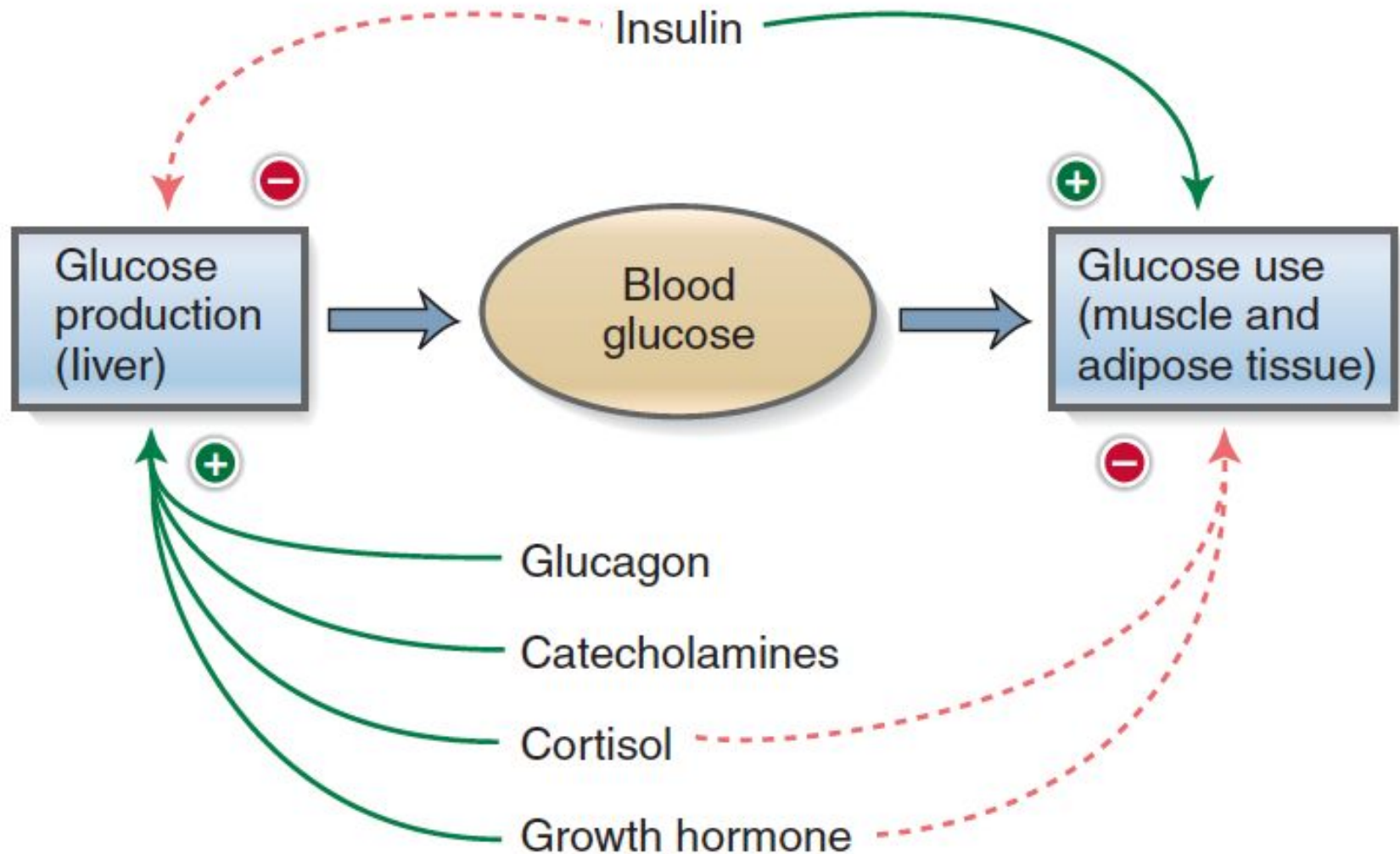
- Plasma glucose is tightly regulated by hormones:

Insulin: ↓ Plasma glucose

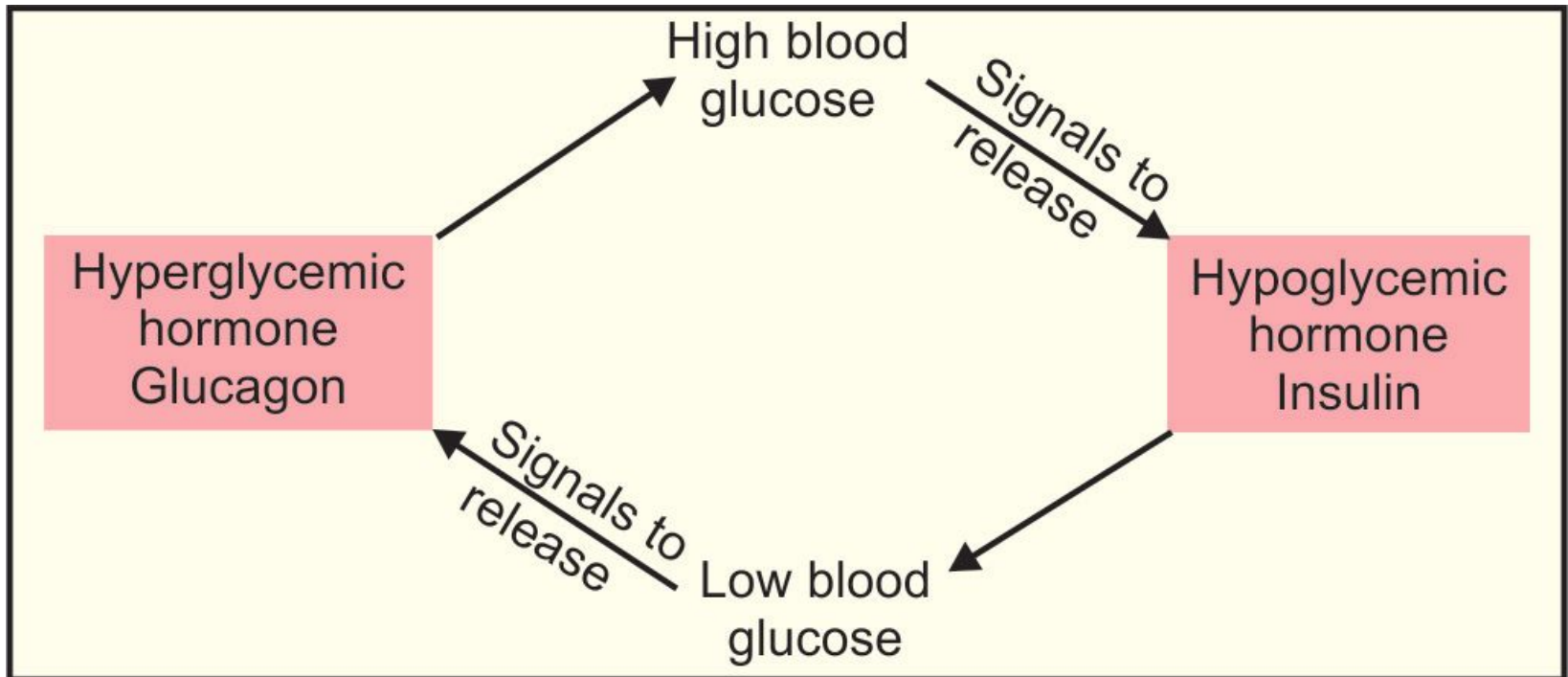
- **Glucagon**
Epinephrine
Cortisol
Growth hormone

↑ Plasma glucose

Integrated Control of Blood Glucose Concentration



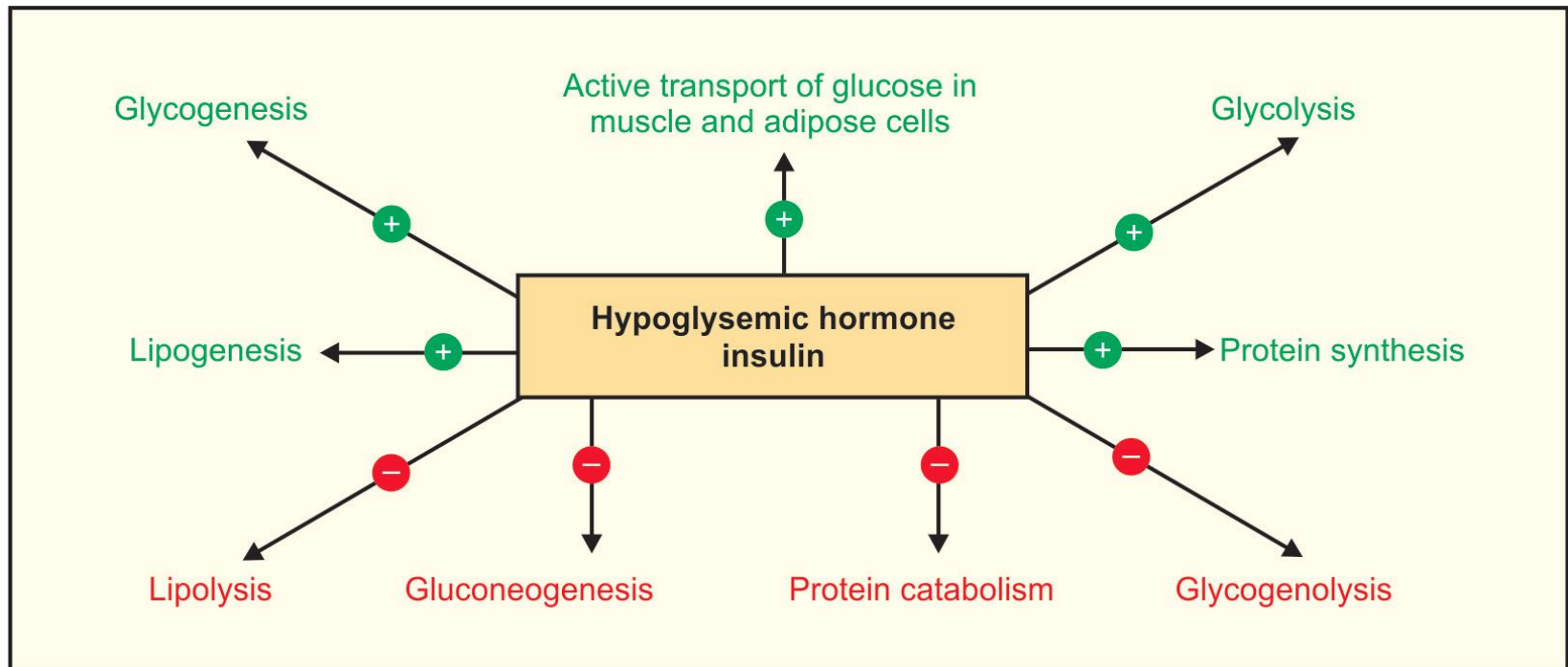
Regulation of blood glucose level by insulin and glucagon



Maintenance of Glucose in Fed State (Hyperglycemic condition)

- Normally, there is an increased blood glucose level shortly after each meal, a postprandial hyperglycemia.
- Increased level of blood glucose releases insulin by β -cells of the Islets of the Langerhans.
 - This hormone reduces the blood glucose level in a number of ways.

Role of insulin in regulation of blood glucose level



Regulation of blood glucose level by insulin

1. By stimulating the active transport of glucose across cell membranes of muscle and adipose tissue but not the liver.
 - Glucose is rapidly taken up into liver as it is freely permeable to glucose.
2. In the liver, increases the use of glucose by inducing:
 - Glucokinase
 - Phosphofructokinase
 - Pyruvate kinase.
3. In the muscle and the liver, stimulates glycogenesis by stimulating glycogen synthase and suppresses glycogenolysis.
4. Insulin inhibits gluconeogenesis by suppressing:
 - Pyruvate carboxylase
 - Phosphoenol pyruvate carboxykinase
 - Fructose 1,6-bisphosphatase
 - Glucose-6-phosphatase.
5. In adipose tissue, stimulates lipogenesis:
 - Glucose is converted to the glycerol-3-phosphate, needed for the formation of triacylglycerol.
6. Increases protein synthesis and decreases protein catabolism.

All these mechanisms are responsible for a drop in blood glucose level.

Insulin Effects on Metabolism

Organ	Pathway	Effect	Enzyme Activity/Metabolite Altered
Liver (100 μ U/ml)	Glucose phosphorylation	↑	Glucokinase
	Glycolysis	↑	Phosphofructokinase-1 (increases concentration of fructose 2,6-bisphosphate), pyruvate kinase, phosphofructokinase-2
	Gluconeogenesis	↓	Phosphoenolpyruvate carboxykinase, fructose 1,6-bisphosphatase, glucose-6-phosphatase, fructose 2,6-bisphosphatase
	Glycogen synthesis	↑	Glycogen synthase
	Glycogenolysis	↓	Glycogen phosphorylase
	Fatty acid synthesis	↑	Acetyl-CoA carboxylase, citrate lyase, and malic enzyme
	Pentose phosphate pathway	↑	Glucose-6-phosphate (G6-P) dehydrogenase
Adipose Tissue	Glucose uptake (@ 10 μ U/ml)	↑	GLUT4*
	Glycolysis	↑	Phosphofructokinase-1
	Pentose phosphate pathway	↑	G6-P dehydrogenase
	Pyruvate oxidation	↑	Pyruvate dehydrogenase
	Triglyceride lipolysis	↑	Lipoprotein lipase
	Fatty acid/triglyceride synthesis	↑	Acetyl-CoA carboxylase
	Lipolysis	↓	Hormone-sensitive lipase
Skeletal Muscle	Glucose uptake (@ 50 μ U/ml)	↑	GLUT4*
	Glycolysis	↑	Phosphofructokinase-1 (increases concentration of fructose 6-phosphate)
	Glycogen synthesis	↑	Glycogen synthase
	Glycogenolysis	↓	Glycogen phosphorylase
	Protein synthesis	↑	Translation (nonselective)

Maintenance of Blood Glucose in Fasting State (Hypoglycemic Condition)

- Decreased level of blood glucose, hypoglycemia causes a release of hyperglycemic hormones, e.g.
 - Glucagon
 - Epinephrine or adrenaline
 - Glucocorticoids
 - Growth hormone and adrenocorticotrophic hormone (ACTH)
 - Thyroxine.

Regulation of blood glucose level by Glucagon

1. Produced by the α -cells of the islets of Langerhans of the pancreas.
2. Opposes the actions of insulin.
3. Acts primarily in the liver as follows:
 - Stimulates glycogenolysis by activating enzyme glycogen phosphorylase and inhibits glycogen synthesis.
 - Unlike epinephrine, glucagon does not have an effect on muscle phosphorylase due to lack of receptors.
4. Enhances gluconeogenesis from amino acids and lactate.
 - Alanine is the predominant amino acid released from muscle to liver (glucose alanine cycle)
 - Lactate formed by oxidation of glucose in skeletal muscle is transported to the liver by lactic acid (Cori cycle).

Glucagon Effects on Metabolism

Organ	Pathway	Effect	Enzyme Activity/Metabolite Altered
Liver	Glycolysis	↓	Phosphofructokinase-1, pyruvate kinase, phosphofructokinase-2
	Gluconeogenesis	↑	Phosphoenolpyruvate carboxykinase, fructose 1,6-bisphosphatase, glucose-6-phosphatase, and fructose 2,6-bisphosphatase
	Glycogen synthesis	↓	Glycogen synthase
	Glycogenolysis	↑	Glycogen phosphorylase
	Fatty acid synthesis	↓	Acetyl-CoA carboxylase
	Fatty acid oxidation	↑	Expression of carnitine palmitoyltransferase 1

Regulation of blood glucose level by Epinephrine or Adrenaline

- Catecholamines secreted by adrenal medulla.
- Stimulates glycogenolysis in the liver and the muscle by stimulating glycogen phosphorylase activity via c-AMP.
 - In muscle (no glucose-6-phosphatase), results in the formation of lactate.
 - In the liver, glucose is the main product, leading to increase in blood glucose.

Epinephrine Effects on Metabolism

Organ	Pathway	Effect	Enzyme Activity/Metabolite Altered
Liver	Glycolysis	↓	Phosphofructokinase-1, decreased F2,6-bisphosphate
	Glycogen synthesis	↓↓	Glycogen synthase
	Gluconeogenesis	↑	Pyruvate kinase, fructose 1,6-bisphosphatase, fructose 2,6-bisphosphatase
	Glycogenolysis	↑↑↑	Glycogen phosphorylase
	Fatty acid synthesis	↓	Acetyl-CoA carboxylase, citrate lyase, and malate dehydrogenase
Adipose Tissue	Lipolysis	↑↑↑	Hormone-sensitive lipase
	Triglyceride uptake from lipoproteins	↓	Lipoprotein lipase
Skeletal muscle	Glycolysis	↑↑↑	Changes concentration of fructose 6-phosphate
	Glycogen synthesis	↓↓	Glycogen synthase
	Glycogenolysis	↑↑↑	Glycogen phosphorylase
	Triglyceride uptake from lipoproteins	↑	Lipoprotein lipase

Regulation of blood glucose level by Glucocorticoids

- Cortisol secreted by adrenal cortex
- Antagonistic to insulin
- Increases:
 - gluconeogenesis.
 - Protein catabolism to provide glucogenic amino acid for gluconeogenesis.
 - Hepatic uptake of amino acids for gluconeogenesis.
- Inhibits:
 - utilization of glucose in extrahepatic tissues.

Regulation of blood glucose level by Growth hormone (GH) and anterior pituitary hormones

- GH and ACTH antagonise the action of insulin.
- GH decreases glucose uptake in the muscle and
- ACTH decreases glucose utilization by the tissue.

Metabolic effects of catabolic hormones

	Catecholamines	Glucagon	Cortisol	Growth hormone
Liver				
Glycogenolysis	↑↑↑	↑↑	None	None
Gluconeogenesis	↑	↑↑↑	↑↑↑	↑
Ketogenesis	↑	↑↑	↑	↑
Muscle				
Glucose uptake	↓	None	↓↓	↓
Ketone metabolism	↓	?	?	?
Adipocyte				
Glucose uptake	↓	None	↓↓	↓
Lipolysis	↑↑↑	None	↑↑	↑

Key: ↑ = stimulatory, ↓ = inhibitory, None = no effect, ? = uncertain.

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Regulation of blood glucose level by Thyroxine

- Secreted by thyroid gland.
- Accelerates hepatic glycogenolysis.
- Increase the rate of absorption of hexoses from the intestine.

Renal Control Mechanism on Blood glucose

- Glucose is continuously filtered by the glomeruli but is normally reabsorbed completely in renal tubules.
 - The capacity of the tubular system to reabsorb glucose is limited.
 - If the blood glucose level is raised above 180 mg/dl, complete tubular reabsorption of glucose does not occur and the extra amount appears in the urine causing glycosuria.
 - 180 mg/dl is the renal threshold value for glucose.
- Helps in regulating the level of glucose in blood:
 - Excrete extra amount of sugar in the urine during hyperglycemic state
 - Reabsorb sugar during the hypoglycemic state

GLYCOSURIA

- Excretion of detectable amount of sugar in urine.
- Results from the rise of blood glucose above its renal threshold level (180mg%).
- Normally, the urine contains about 0.05gm% of sugar.
 - Cannot be detected by Benedict's test
 - Under certain circumstances, a considerable amount of glucose or other sugar may be excreted in the urine.
- Classified into:
 1. Alimentary (lag storage) glycosuria:
 - Rise of blood sugar rapidly after meal above the normal renal threshold (180mg%)
 2. Renal glycosuria:
 - Due to impaired tubular reabsorption of glucose
 - Have lowered renal threshold (may be 130-150mg%)
 3. Diabetic glycosuria:
 - Due to deficiency or lack of insulin
 - Blood glucose level exceeds the renal threshold

CLINICAL CORRELATES

Decreased production of insulin, which is usually caused by autoimmune destruction of pancreatic β cells, results in **type 1** (formerly called insulin-dependent) diabetes mellitus. Type 1 diabetes is characterized by **hyperglycemia**, the result of decreased uptake of glucose by cells and increased output of glucose by the liver (due to low insulin and high glucagon levels in the blood). These patients are dependent on exogenous insulin to survive.

CLINICAL CORRELATES

Decreased release of insulin from the pancreas or **decreased sensitivity** of tissues to insulin (insulin resistance) results in **type 2** (formerly called non-insulin-dependent) **diabetes mellitus**. This condition also is characterized by **hyperglycemia**.

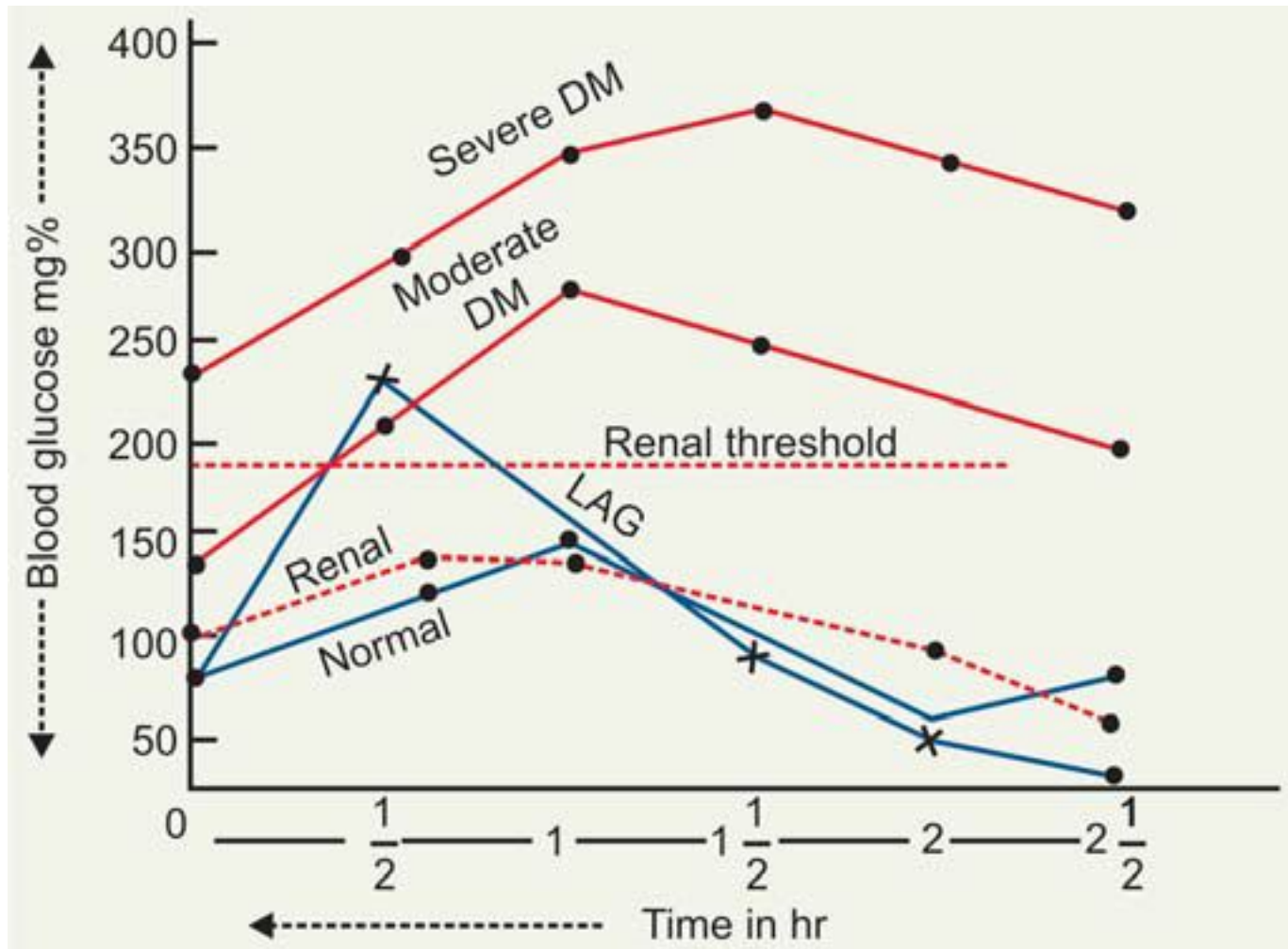
CLINICAL CORRELATES

The **oral glucose tolerance test** is one test used to diagnose diabetes. Patients drink a liquid containing 75 g of glucose dissolved in water. **After 2 hours**, the serum glucose is measured. A blood glucose of **less than 139 mg/dL is normal**, whereas a level of **140 to 199 mg/dL represents "prediabetes"** or impaired glucose tolerance. A serum level of **more than 200 mg/dL is indicative of diabetes**.

CLINICAL CORRELATES

Diabetes is more often diagnosed based on **fasting plasma glucose tests**. Patients are required to **fast for at least 8 hours**, and the tests are most reliable in the morning. Normally, the serum glucose should be less than 100 mg/dL; a blood glucose, on **at least two separate occasions, of more than 126 mg/dL indicates diabetes**, whereas a value **100 to 125 mg/dL indicates impaired glucose tolerance (prediabetes)**.

Different glucose tolerance curves



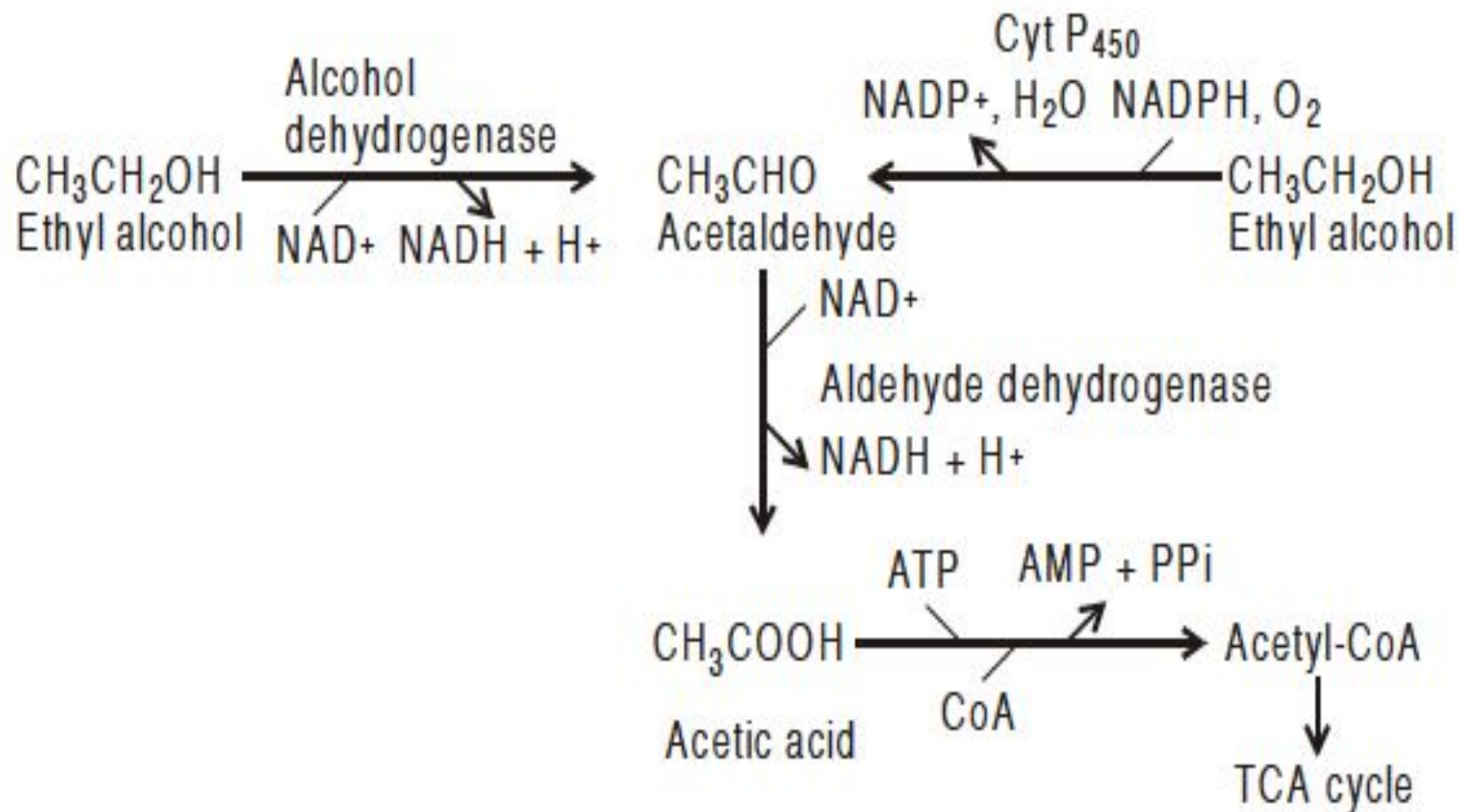
Alcohol Metabolism and Gluconeogenesis

Dr Balakrishnan S, PhD.,

BioMedical significance of Alcohol Metabolism

1. It serves as energy source like carbohydrate. However, if consumed excess (chronic alcoholism) leads to fatty liver development.
2. Some individuals (chronic alcoholics) develop vit A and thiamin deficiency.
3. Lactic acidemia occurs due to excess NADH in the cytosol.
4. Moderate consumption of alcohol was found to be beneficial to stroke patients.
 - HDL level is higher in people consuming small amount of alcohol daily.
5. Clearance of toxins from circulation by liver becomes slow and some compounds get converted to carcinogens in chronic alcoholics.
6. Alcohol consumption aggravates gout and porphyrias.

ALCOHOL METABOLISM



ALCOHOL METABOLISM

Major pathway:

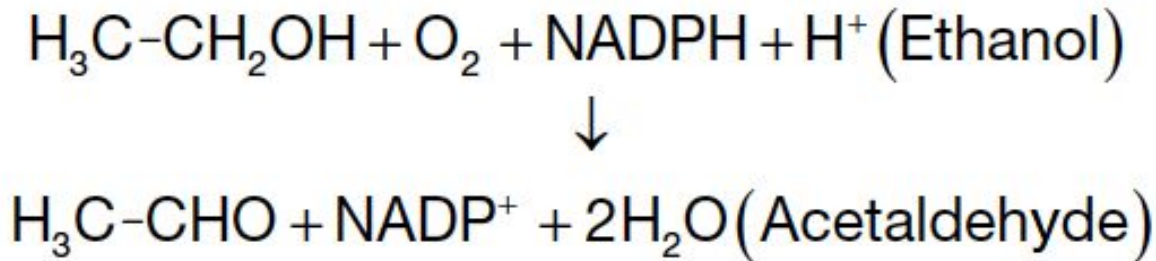
- Alcohol cannot be excreted and must be metabolized, primarily by the liver.
- This metabolism comprises two steps:
 - The first step, catalyzed by the enzyme *alcohol dehydrogenase*, takes place in the cytoplasm
 - The second step, catalyzed by *aldehyde dehydrogenase*, takes place in mitochondria.
- Acetate, once produced, is not toxic, so a slow-acting acetyl-CoA synthetase (which uses acetate as a substrate) converts into acetyl-CoA .



ALCOHOL METABOLISM (CONT'D)

Minor pathway:

- Some alcohol is metabolized by the cytochrome P-450 system:

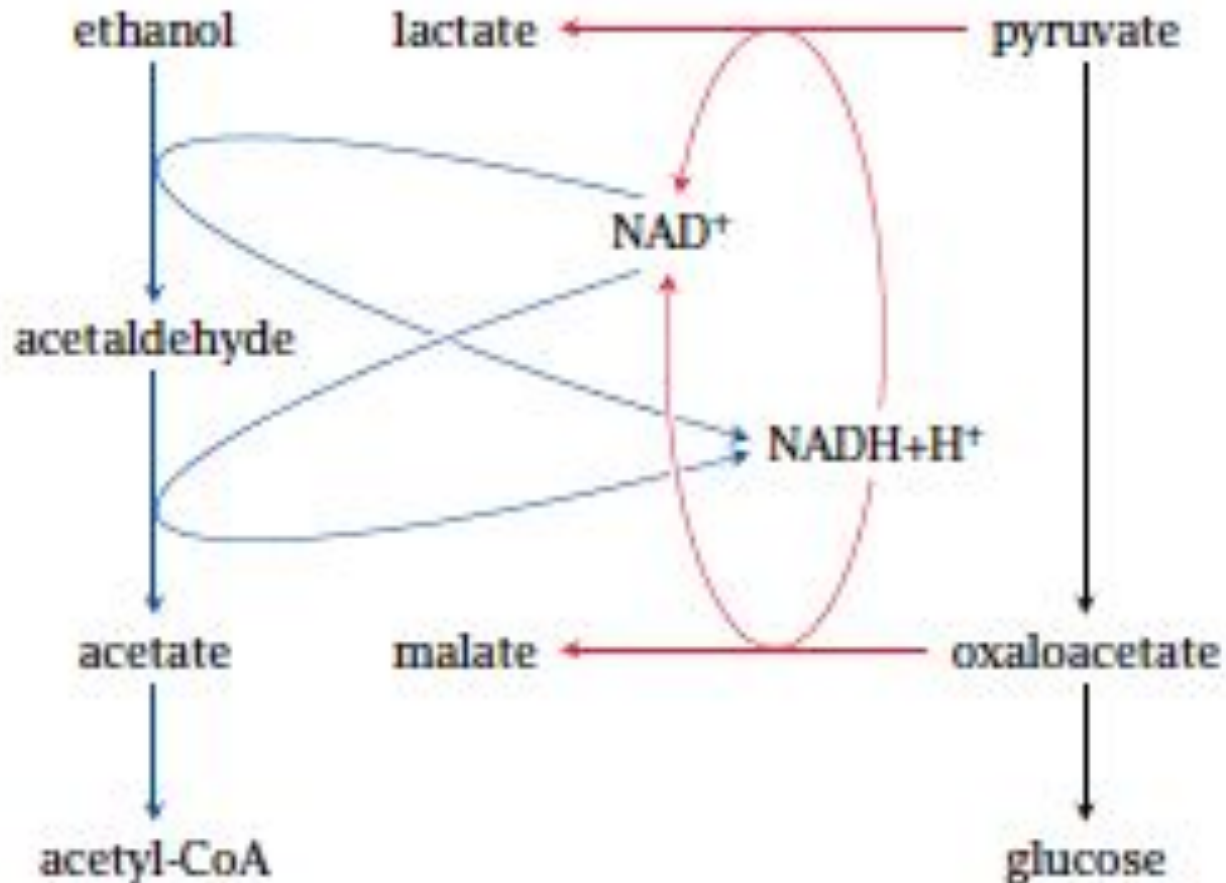


- Most people metabolize about 10 g of alcohol per hour, and the blood alcohol level decreases by about 0.15 g/L every hour.
 - These calculations are important when a blood sample from a drunken driver has been obtained some hours after an accident.

Alcohol metabolism Vs Gluconeogenesis

- Note that Alcohol consumption leads to an accumulation of NADH.
- This **high conc. of NADH inhibits Gluconeogenesis**
- In fact, high conc. of NADH will cause the reverse reaction to predominate.
- High NADH favors the formation of:
 - Lactate from pyruvate
 - Malate from OAA in the cytoplasm
 - Glycerol-3-phosphate from DHAP
- The consequences may be **hypoglycemia and lactic acidosis.**

Ethanol degradation inhibits gluconeogenesis



Alcohol metabolism Vs Fatty acids (FA) oxidation

- The **NADH** glut also inhibits **FA oxidation**.
- The **purpose of FA oxidation is to generate NADH for ATP** by oxidative phosphorylation, but an alcohol consumer's NADH needs are met by ethanol metabolism.
- In fact, the excess NADH signals for FA synthesis. Hence, TG accumulates in liver, may lead to **Fatty liver**.

Effects of alcohol on liver metabolism

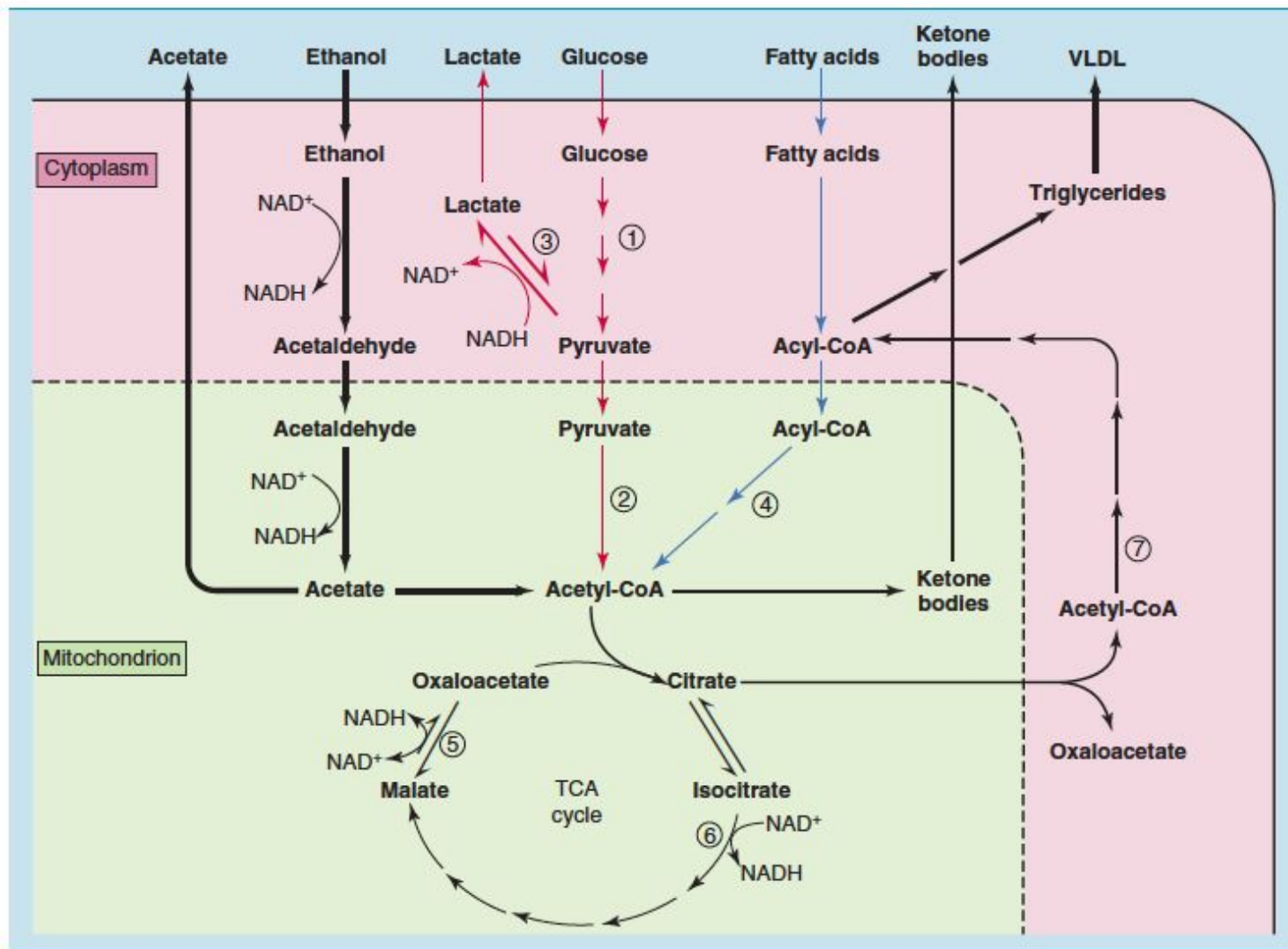


Fig. 32.24 Effects of alcohol on liver metabolism. Important control points: ① phosphofructokinase: inhibited by high energy charge; ② pyruvate dehydrogenase: inhibited by high energy charge, high NADH/NAD⁺ ratio, and high acetyl-coenzyme A (CoA); ③ lactate dehydrogenase: high NADH/NAD⁺ ratio favors lactate formation; ④ β-oxidation: slowed down because of low NAD⁺; ⑤ malate dehydrogenase: high NADH/NAD⁺ ratio favors malate formation, oxaloacetate is depleted; ⑥ isocitrate dehydrogenase: inhibited by high energy charge and high NADH/NAD⁺ ratio; and ⑦ acetyl-CoA carboxylase: may be stimulated by high citrate (accumulates because isocitrate dehydrogenase is inhibited). TCA, Tricarboxylic acid; VLDL, very-low-density lipoprotein.

Read more...



THANK YOU...